

IES Possibilia Preview Release

Briefing pack v0.1 (Dec 2025)

Introduction

This document provides a briefing on a preview release of IES Possibilia - an RDF implementation of early-stage ideas for representing uncertainty within a 4D ontology and which is built upon [ies-top](#) and [ies-core](#). These ideas were first introduced at STIDS 2024¹.

This implementation is one possible approach among many of realizing the ideas presented at STIDS - some of which may ultimately prove more effective for addressing the requirements identified in that paper. We are sharing this preview to give interested parties an early sense of the ontology's emerging direction: the topics we are investigating, the aspects that still require refinement, and the challenges we are currently working to resolve - one of which is illustrated in Appendix A.

Readers will notice that significant development remains ahead. Nevertheless, we hope this preview serves as a useful guide to the kinds of philosophical commitments and design considerations involved in representing uncertainty within a 4D ontology.

1. [Partridge, C., Mitchell, A., & Cola, A. \(2025\). *Digitalizing Uncertain Information*](#)

Naming of RDF resources

ies-possibilia is grounded in the philosophical topics of Possible Worlds, Counterpart Theory, Modality and *De Se* ascription. Translating these into a physical implementation (like RDF) inevitably requires making choices - whether due to limitations of the chosen implementation technology or for user experience reasons.

Given the early stage of this work, we have chosen to use intentionally opinionated RDF resource names (classes and properties) that remain as close as possible to the terminology of the underlying philosophical works. These names are highlighted within purple bordered outlines.

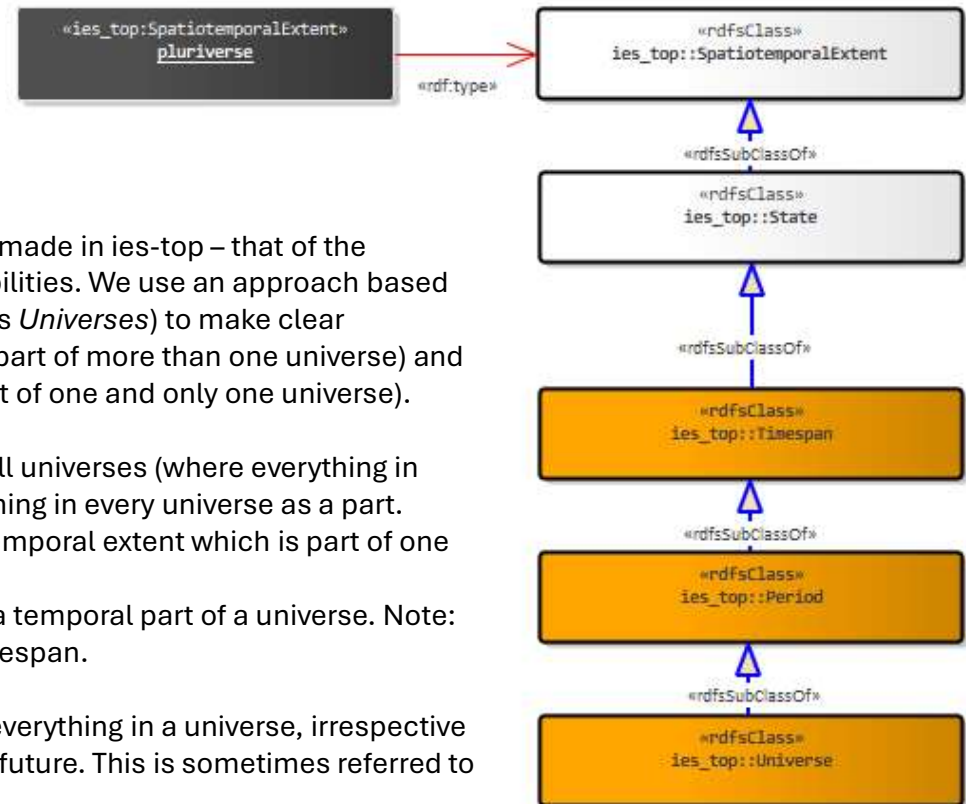
At the same time, we have provided alternative naming suggestions that may feel more familiar and accessible to practitioners. These are found within green bordered outlines.

NOTE: Throughout this document, RDF resources referenced from *ies-top* and *ies-core* are identified using their respective prefixes. Newly introduced RDF resources belonging to *ies-possibilia* appear without any prefix.

A reminder from ies-top: The Pluriverse

Before introducing IES-Possibilia, we first revisit the related commitment made in ies-top – that of the *Pluriverse* – which provides the necessary grounding for talk about possibilities. We use an approach based on David Lewis' interpretation of *Possible Worlds*² (referred to in ies-top as *Universes*) to make clear distinctions between spatiotemporal extents that can be trans-universe (part of more than one universe) and the ones we do most of our work with, those that are universe-bound (part of one and only one universe).

- **pluriverse** - An instance of SpatiotemporalExtent which is the sum of all universes (where everything in each universe is part of that universe). Put another way, this has everything in every universe as a part.
- **State** – an spatiotemporal extent that is universe-bound i.e. an spatiotemporal extent which is part of one and only one universe.
- **Timespan** – a state (i.e. universe-bound spatiotemporal extent) that is a temporal part of a universe. Note: a universe is an improper temporal part of itself – and so a maximal timespan.
- **Period** – a connected and uninterrupted timespan.
- **Universe** – a maximal spatiotemporal connected extent that includes everything in a universe, irrespective of any indexing, such as whether it is present now, in the past, or in the future. This is sometimes referred to as a world.



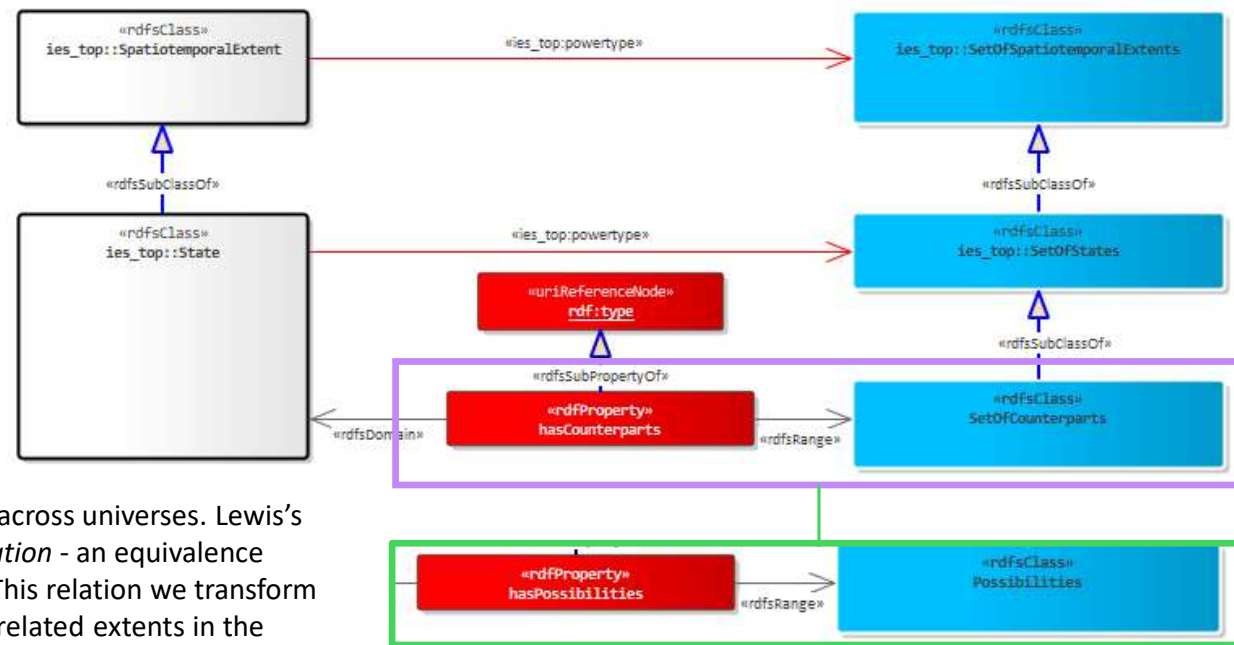
Introducing IES-Possibilia

Possibilities

Because individuals such as a person, a vehicle, a location and an event are **States** (spatiotemporal extents bound to a single universe), discussing possibilities involving them - for example, the possibility of a person called Anne being in Edinburgh at a given time - implies, under the Possible Worlds approach, the existence of two distinct and unconnected extents: a possible Anne who was in Edinburgh and a possible Anne who was not. These must be distinct, since it is impossible to both be and not be in Edinburgh at the same time. And given they are universe-bound, then they must be in different universes.

We therefore need a means to connect these distinct extents across universes. Lewis's answer to this was to connect them using the *counterpart relation* - an equivalence relation that links individuals who are in some sense, similar. This relation we transform into a set (SetOfCounterparts) containing all counterpart-related extents in the pluriverse. Each set is one counterpart relation and therefore contains as members all extents that are related as counterparts. For example, all the possible ways Anne could have lived her entire life, all the possible Edinburghs, or all the possible states of Anne at a given time.

We could consider this SetOfCounterparts, a set of Possibilities (see green outlined box) where we can associate members to this set using a special form of rdf:type- hasCounterparts / hasPossibilities.



Combined Possibilities (1 of 2)

In most cases we do not speak of possibilities in isolation. For example, if Anne was in Edinburgh on the 1st January 2025 - then this possibility involves at least four spatiotemporal extents:

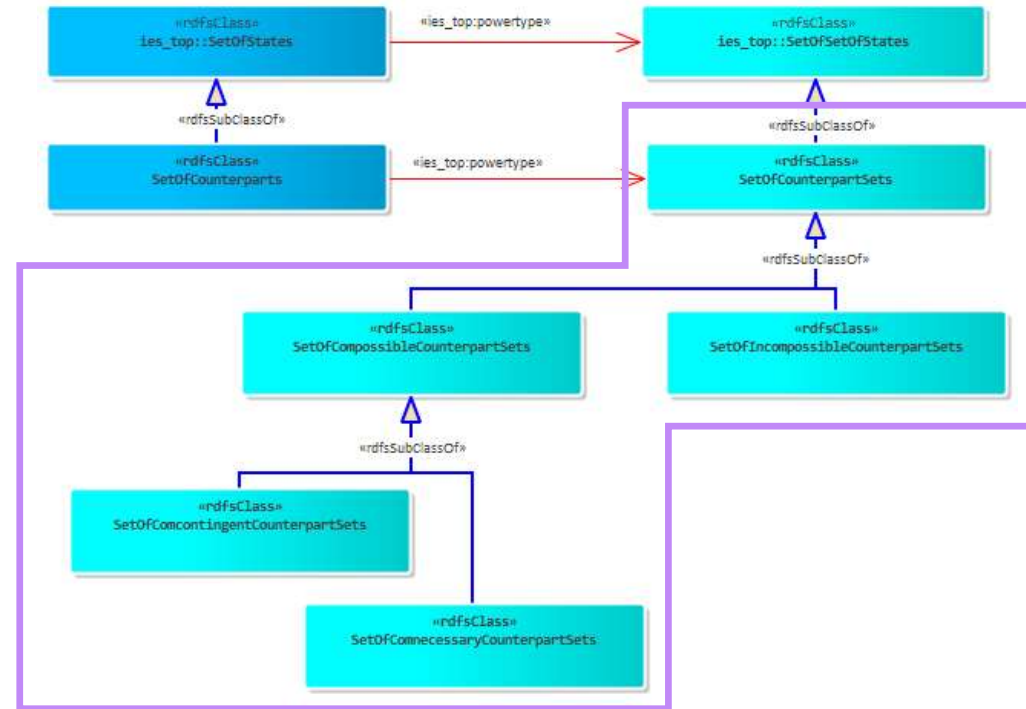
1. Anne;
2. Edinburgh;
3. the period of time - 1st January 2025;
4. the temporal part of Anne at Edinburgh on the 1st January 2025.

There must be a universe where all four exist together – that is, a universe where they are jointly possible. Consequently, we need a means to talk about *combined possibilities*. This will become especially important when we turn to the topic of belief, which we will discuss later.

Compossibility is a philosophical concept from Gottfried Wilhelm Leibniz which helps us deal with combined possibilities. Leibniz said that a possible world is made up of individuals that are all compossible with one another. He also said that the existence of one individual may negate the possibility of the existence of another i.e. they are impossible.

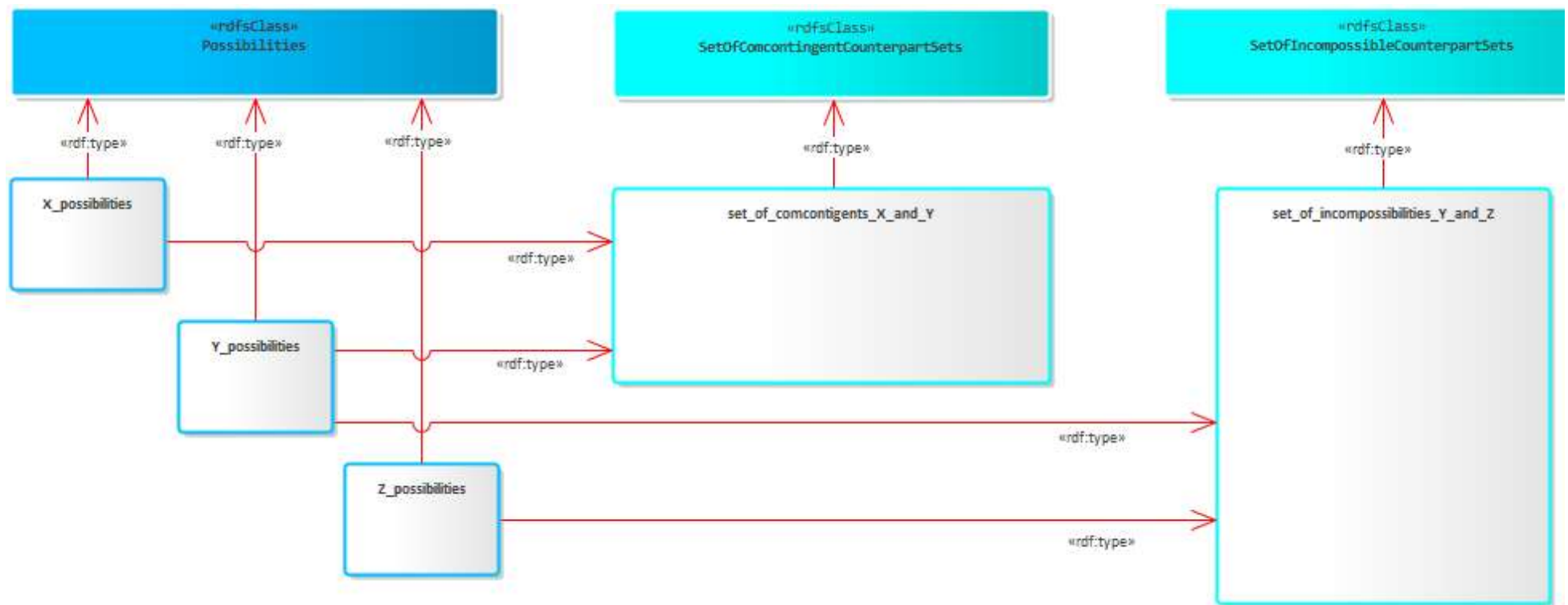
We take from Leibniz the concept of Compossibility to introduce a means of grouping Possibilities (SetOfCounterparts) into sets that are:

- **Compossible** - two or more counterpart sets that are jointly possible - there is at least one universe which includes counterparts from each set.
- **Comcontingent** - two or more counterpart sets that are jointly contingent - there is at least one universe, but not all, which includes counterparts from each set.
- **Comnecessary** – two or more counterpart sets that are jointly necessary - all universes include counterparts from each set
- **Impossible** - two or more counterpart sets that are jointly impossible or incompatible with one another - there are no universes which include counterparts from each set.



Combined Possibilities (2 of 2)

Here we provide an example illustrating the use of the newly introduced RDF classes. In this example, X and Y are stated to be *comcontingent*, whereas Y and Z are *impossible*.



Belief (1 of 4)

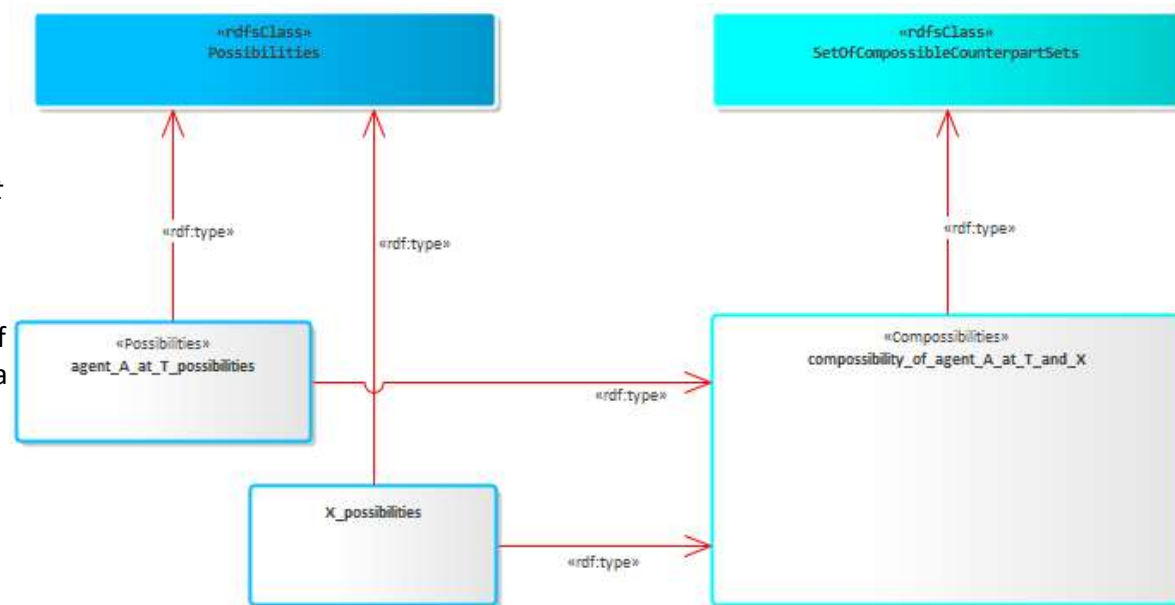
So far, we have a means of talking about possibilities and combined possibilities, but only from a God-like, agent-agnostic perspective. However, possibilities belong to some thing - they are beliefs held by an agent, whether that is a person, an organization, information in a computer system or even the writings on a piece of paper. Moreover, they belong to an agent *at a time*, since agents can change their beliefs.

Our solution here is to identify the subsets of `SetOfCompossibleCounterpartSets` which include as one of its members, the possibilities (set of counterparts) of an agent at a time. This subset can then be used as a conduit for that agent's time-indexed beliefs. For example (see figure), when an agent *A*, at time *T*, believes in the possibility of *X*, then there is a compossible set which includes two members:

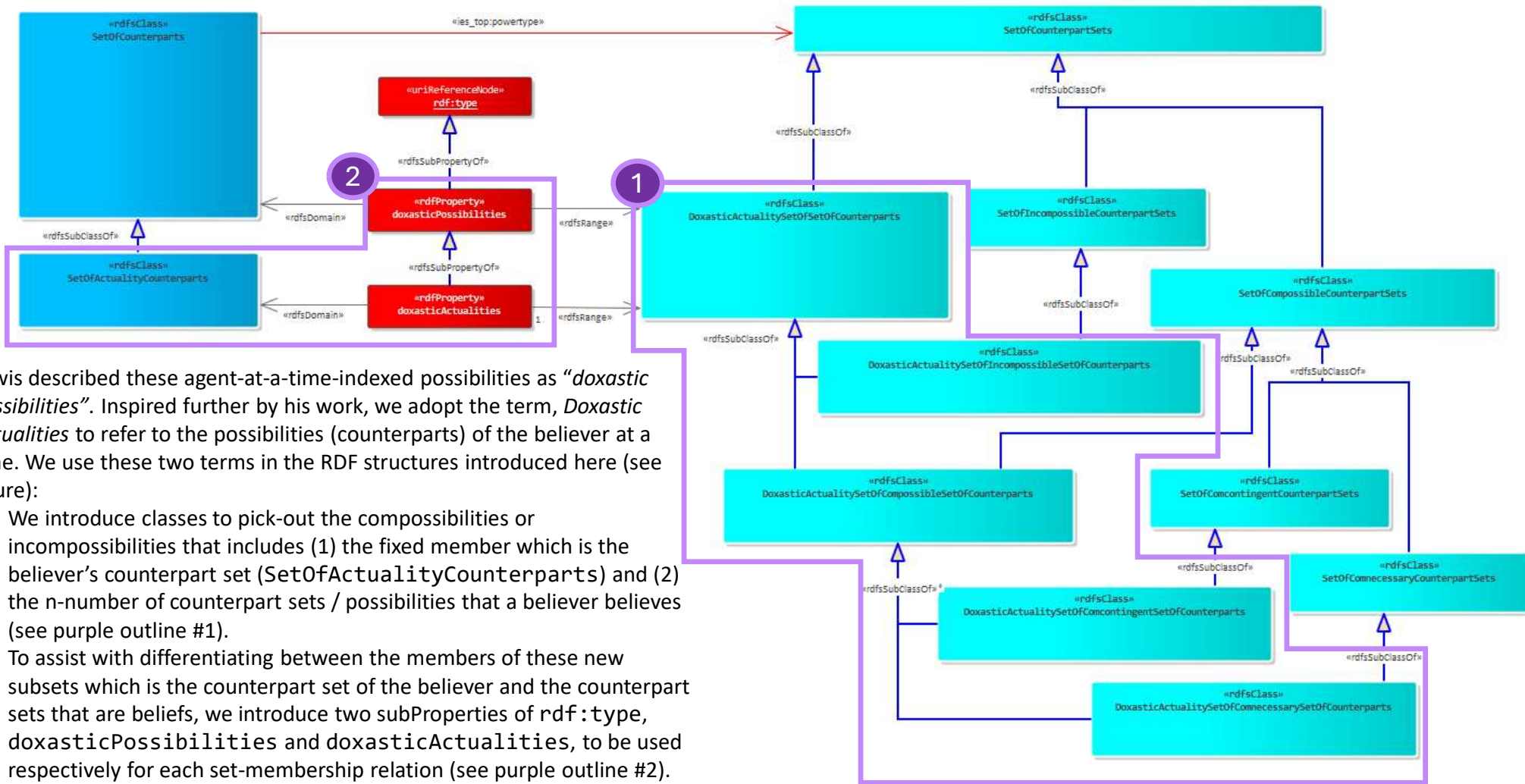
1. The possibilities of agent *A* at a time *T* i.e., the set of counterparts of *A* at *T*, who hold the same beliefs;
2. The possibilities of *X* i.e., the set of counterparts of *X*.

Within this subset of compossibilities, one member remains fixed - the possibilities of an agent at a particular time. In this sense, the subset functions as a structured container for agent-centered beliefs.

The next slide will show how we implement this subset of Compossibility (and similarly for Impossibility) in our RDF.



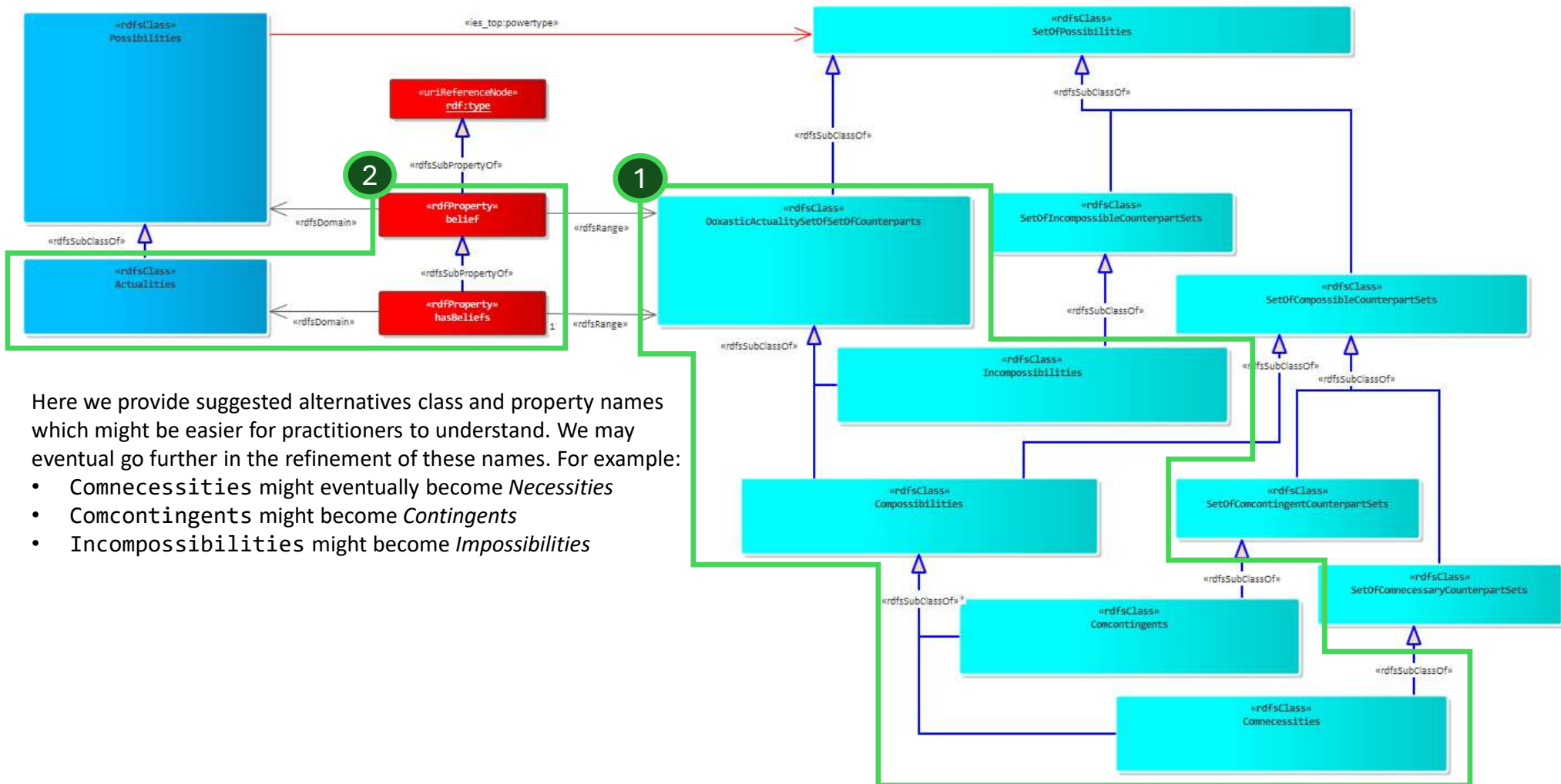
Belief (2 of 4)



Lewis described these agent-at-a-time-indexed possibilities as “*doxastic possibilities*”. Inspired further by his work, we adopt the term, *Doxastic Actualities* to refer to the possibilities (counterparts) of the believer at a time. We use these two terms in the RDF structures introduced here (see figure):

- We introduce classes to pick-out the compossibilities or impossibilities that includes (1) the fixed member which is the believer's counterpart set (`SetOfActualityCounterparts`) and (2) the n-number of counterpart sets / possibilities that a believer believes (see purple outline #1).
- To assist with differentiating between the members of these new subsets which is the counterpart set of the believer and the counterpart sets that are beliefs, we introduce two subProperties of `rdf:type`, `doxasticPossibilities` and `doxasticActualities`, to be used respectively for each set-membership relation (see purple outline #2).

Belief (3 of 4)

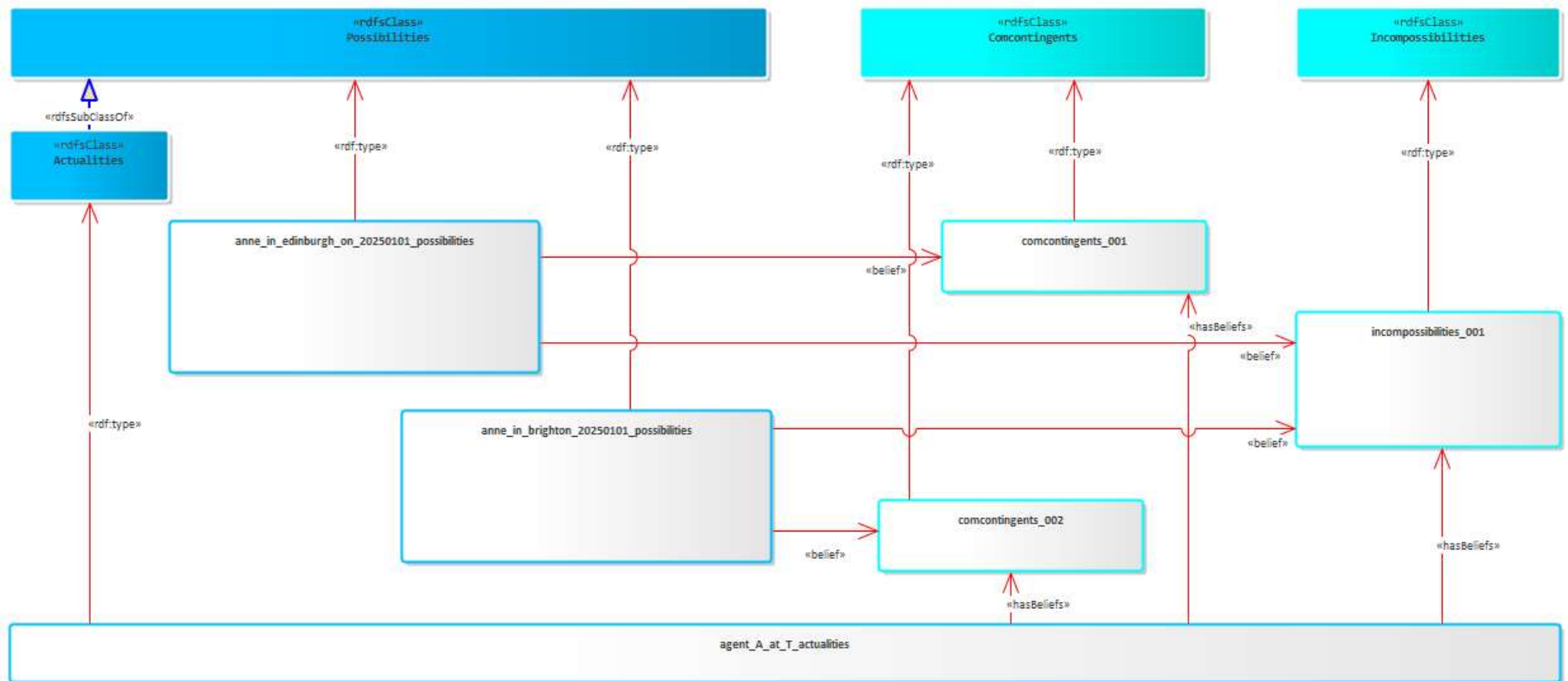


Here we provide suggested alternatives class and property names which might be easier for practitioners to understand. We may eventual go further in the refinement of these names. For example:

- `Connecessities` might eventually become *Necessities*
- `Comcontingents` might become *Contingents*
- `Impossibilities` might become *Impossibilities*

Belief (4 of 4)

Here we provide an example of how these RDF structures can be used. In this example, agent A at time T believes that Anne, on the 1st January 2025, could have been (contingently) either in Edinburgh (comcontingents_001) or Brighton on the 1st January 2025 (comcontingents_002). However, agent A also believes that she could not have been in both places at the same time (incompossibilities_001).

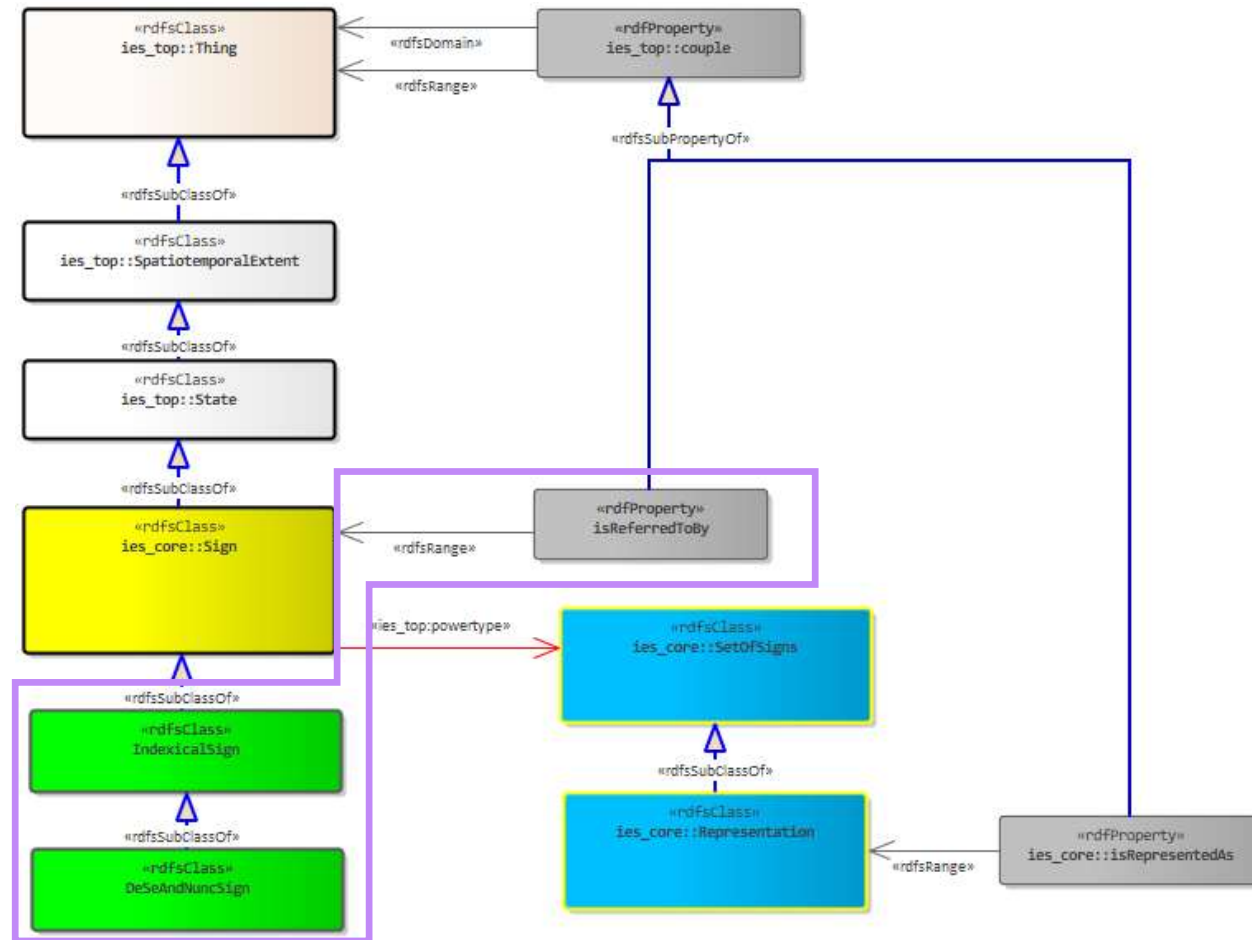


Me and Now (1 of 2)

The possibilities represented by a bearer - whether a database, a digital or physical file, or a written record - may originate from multiple sources. Moreover, these possibilities are not always current; they change over time.

We need a means to give any information bearer the ability to identify both the possibilities *it* owns, and which are *current* (i.e., the possibilities that are valid *now*). This requires making explicit within the information, two indexicals: “*me*” – denoting the system’s self-identity (referred to in philosophy as the *de se*) and “*now*” (philosophical termed the *de nunc*)^{3, 4}.

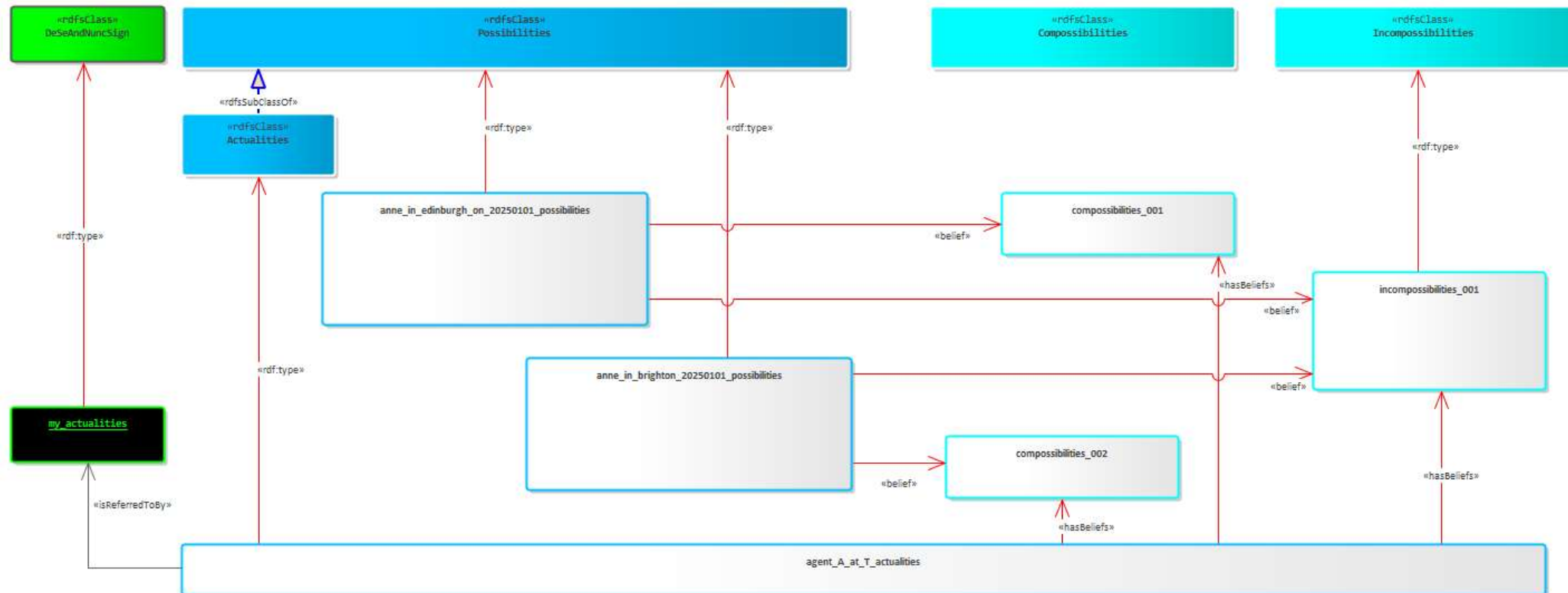
In our RDF implementation, we achieve this by introducing `IndexicalSign` which extends `ies_core:Sign`. A dedicated subset: `CurrentMeSign`, is then used to explicitly reference a me’s actuality (its counterpart set) using a newly introduced `isReferredToBy` couple – see the example on the next slide.



3. C. Partridge, M. Khan, S. de Cesare, F. Gailly, M. Verdonck, and A. Mitchell, 'Thoroughly Modern Accounting: Shifting to a De Re Conceptual Pattern for Debits and Credits'
4. C. Partridge, S. deCesare, A. Mitchell, A. Leon, F. Gailly, and M. Khan, 'Ontology then Agentology'

Me and Now (2 of 2)

Here we build upon the example concerning the possibilities of Anne being in Edinburgh or Brighton on 1st January 2025. We retain the contingent possibilities already ascribed - that Anne was in Edinburgh or Brighton, but not both - as beliefs held by agent *A*. We then mark the actuality counterpart set, `agent_A_at_t_actualities` with a `DeSeAndNuncSign`, making explicit in our data that `agent_A_at_t_actualities` is the counterpart set of the information store at time *T*, where *T*, is now.



Appendix 1: Challenge – set-level articulation of relations

Here we illustrate the complexity involved in describing relations at the *counterpart set* or *possibility* level. Unlike single instances - such as individual people, locations, states, or time periods - whose relationships can be straightforwardly expressed using RDF predicates, expressing the equivalent relationships at the set-level is far more challenging.

Because there may be multiple counterparts of “Anne in Edinburgh on 1 January 2025” across the Pluriverse, any relevant relations (e.g., to Anne, to Edinburgh, to the 1 January 2025) must apply to all members of the counterpart set (*anne_in_Edinburgh_on20250101_possibilities*). Specifically, every instance of this set must bear an *isTemporalPartOf* relation to Anne and an *isPartOf* relation to both Edinburgh and the period of time, 1 January 2025.

Normally we only need to handle these sorts of semantics at the ontology-level i.e., by instantiating instances of RDF properties and using RDFS Domain and Range to prescribe the source and target of such relations. However, because there is no logical distinction between the sets defined in the ontology and those present in our data, we have explored applying this same approach here as well. As you can see it is a costly approach. This area clearly requires further investigation and innovation.

